e-values for multiplicity control in multiverse analysis

Paul Rognon-Vael[1] and David Rossell[2,3*]

[1*]Department of Economics and Business, Bocconi University, Via Roberto Sarfatti, 25, Milano, 20136, Italy.
[2*]Department of Economics and Business, Universitat Pompeu Fabra, Ramón Trias Fargas 25, Barcelona, 08005, Spain.
[3]Data Science Center, Barcelona School of Economics, Ramón Trias Fargas 25, Barcelona, 08005, Spain.

*Corresponding author(s). E-mail(s): david.rossell@upf.edu;
Contributing authors: paul.rognon@gmail.com;

Abstract

Multiverse analysis refers to a common situation where one wishes to assess the association between multiple possible treatment definitions and multiple possible outcome definitions, potentially within multiple sub-populations, among other possible analysis specifications. Multiverse analysis is a useful exploratory tool to assess heterogeneity across the considered specifications, but it is sometimes also used to assess statistical significance. Then, it is critical to acknowledge that multiple comparisons are being performed, and to ensure a valid statistical control of false positive findings. We study the use of e-values within generalized linear models as a tool to control the false discovery rate regardless of the dependence structure of the multiple analyses being performed, while accounting for confounding covariates. We compare via simulations the performance of several approaches: universal e-values, soft-rank e-values, and p-to-e calibration. We find that, for problem characteristics typically encountered in multiverse analyses, a mixture-based p-to-e calibration significantly outperforms the other two approaches in terms of statistical power, but said power may be moderate unless the sample size or effect sizes are large enough. An application studying the association between teenager technology use and mental well-being reveals association between depression, low self-steem and peer problems with internet and social media usage.

Keywords: multiverse analysis, e-value, multiple testing, false discovery rate

1 Introduction

There are many applications where one wants to assess the effect of multiple treatments on multiple outcomes, potentially also separately within multiple sub-populations. A prominent example in the Social Sciences occurs when one has multiple outcomes that can be conceivably thought of as measuring related underlying characteristics, such as different measures of psychological well-being, several measures of economic activity, or several measures of radicalization in social media usage. Similarly, the multiple treatments may refer to using various types of technology, realizing various economic interventions, or exposure to various social media unradicalization campaigns. It may also be natural to wish to assess treatment effects within sub-populations, for example defined by gender and age groups, economic regions, or social media user backgrounds. Beyond the Social Sciences, similar situations arise in other fields. In Biomedicine one may study the effect of non-identical, but related, treatments on multiple measures of disease progression, in Climate Science one may study the association between multiple human-driven processes on multiple physical parameters, etc.

Multiverse analysis provides a data analysis framework that considers such multiple outcome-treatment-subpopulation combinations. A key pitfall that it seeks to avoid is selective reporting, whereby researchers consider all treatment-outcome-subpopulation combinations and report only those where the estimated effects are larger or more statistically significant. Critically, selective reporting occurs when a researcher fails to indicate all the combinations that were considered. By acknowledging all such combinations, multiverse analysis serves as an excellent tool as an exploratory data analysis method that assesses the heterogeneity or the sensitivy of results across various analysis specifications. In some instances however, researchers may be interested in going beyond a descriptive analysis to also report some assessment of statistical significance. In such circumstances one may wish to control some measure of false discoveries, and if so it is then critical to account for the multiple comparisons when assessing the statistical significance of the findings. This is the strategy followed in this paper. To be clear, we do not argue that hypothesis testing should be preferred over descriptive analyses or confidence intervals, simply that if one wishes to test hypotheses then this should be done properly.

Before proceeding, we discuss another popular strategy: focus the analysis on some summary measure across all the comparisons being made. This is the standard approach for meta-analyses in the medical literature, where one combines the evidence regarding the effect of a medical treatment (or family or related treatments) across studies (typically, randomized clinical trials). Therein, it is common to account for treatment effect heterogeneity, often driven by differences in the patient population targeted by each study. A notable example in the Social Sciences, where randomized trials are rare, is Specification Curve Analysis (SCA, [1]). By default, SCA reports the median estimated effect across treatment-outcome-subpopulations, and assesses the statistical significance of this single parameter. An advantage of using a single summary measure over multiple comparison adjustments is that, since no significance corrections are needed, if all treatment effects are comparable then one may have higher statistical power to detect that the summarized effect truly exists. However, we urge

strong caution in the sense that aggregate measures should only be used in appropriate circumstances. A critical pitfall is that, when the treatments are not comparable or there is significant heterogeneity in their effects, the aggregate measure may be highly misleading. For example, [2] showed that SCA led [3] to erroneously concluding that the association between technology use and teenager well-being was practically negligible. The reason was that the association was near-zero for some technologies such as television use, but large for others such as mobile phone use, and hence overall mild on the average. Another critical issue arises when the aggregate measure is obtained across multiple possible subsets of control covariates, an option that is explicitly considered in SCA. This is inadequate because one combines quantities that have a different statistical meaning, such as marginal and conditional associations, and hence the summary has no natural interpretation. For example, assessing the conditional association between salary and gender given an individual's job position, education and other characteristics has a different meaning than assessing the marginal association between salary and gender. Namely, a marginal association between gender and salary can be partially driven by people from different genders making different career choices, whereas an association between gender and salary within people with the same job position, education and characteristics suggests a possible salary discrimination. See [2] for an in-depth discussion of issues with SCA.

In this paper we study the use of a simple approach to multiplicity control in multiverse analysis based on e-values. Briefly, e-values are an increasingly popular framework for hypothesis testing that has several features relative to p-values. When comparing two hypotheses, e-values accumulate evidence against the null hypothesis similarly to a Bayes factor (see [4] for a review). Further, e-values guarantee a strict control of the type I error probability, and are particularly suited to control the false discovery rate (FDR) in multiple comparisons [5]. Although less relevant to this work, e-values also enable anytime-valid inference, a strong asset in settings where one accumulates data and makes decisions sequentially. For seminal papers on e-values see [6–8], and for an excellent monograph see [9].

We briefly review the rich literature on FDR control for multiple testing. [10] proposed that, when conducting many statistical tests, one should use procedures that guarantee controlling the FDR below a user pre-specified level α. The FDR is the expected proportion of false positives among the hypotheses that are claimed as statistically significant. [10] proposed a p-value adjustment procedure that controls the FDR under the assumption that the tests are independent, and [11] a variation that assumes that the tests follow a type of positive dependence structure. A large body of subsequent work extended the FDR framework in several directions. A prominent example are the q-values of [12]. Analogously to p-values controlling the expected number of false discoveries, q-values control the expected positive FDR, that is the expected fraction of false discoveries, conditional on ≥ 1 discoveries being reported. However, q-values require estimating nuisance parameters such as the proportion of true null hypotheses, and therefore do not control the FDR exactly. Further, estimating said nuisance parameters often requires simplifying assumptions such as independence between the test statistics, which rarely hold in practice. Another influential line of work is based on the local FDR [13], which is embedded into an empirical Bayes

framework. The local FDR can be interpreted as the posterior probability that a given hypothesis is null, given the value of the test statistic for that hypothesis. The local FDR provides a more refined, hypothesis-specific, assessment of significance than tail-area measures such as q-values. Estimating the local FDR accurately requires the number of hypotheses being tested to be larger than is common in multiverse analyses (at least in the hundreds), assuming independence across tests, and making certain choices when estimating the global mixture distribution of the test statistics. Further related work includes the optimal discovery procedure of [14], and the Bayesian discovery procedure of [15]. See [16] for a monograph on FDR control.

An advantage of e-values over these methods are that they provide a finite-n FDR control that does not require estimating nuisance parameters, and which holds regardless of the number of tests being performed and of the dependence structure between the multiple tests. This is relevant in multiverse analysis, where typically tests cannot be assumed independent. Another advantage of e-values for multiverse analysis is that they are more composable, for example they can be multiplied or averaged across independent datasets (corresponding to subpopulations where a treatment effect is tested, for example) to obtain valid e-values. A main limitation of e-values relative to methods based on p-values, q-values and empirical Bayes is that the latter usually have a higher statistical power, provided that their assumptions hold. Therefore, e-values are most appropriate for datasets where one has relatively large sample sizes. Otherwise, while FDR control is ensured, one may obtain few statistically significant findings. In this paper, we consider several strategies to obtain e-values and compare their power via simulation studies that mimic common sample sizes and problem dimensions encountered in multiverse analysis within the Social Sciences. Our findings suggest that simple p-to-e calibration rules, which convert a p-value into an e-value, attain higher statistical power than generic strategies such as universal e-values [17] and soft-rank e-values [18].

The paper is structured as follows. In Section 2 we introduce notation and lay out a generalized linear model framework for treatment effect estimation within the context of a multiverse analysis. Section 3 reviews e-values, discusses their use for multiplicity control, and assesses the power of several e-value methods via linear and logistic regression simulation exercises. Section 5 applies our methodology to a dataset studying the association between teenager mental well-being and technology use. Section 6 concludes.

2 Framework

We lay out a multiverse framework that is based on generalized linear models and closely resembles that of [2] but provides FDR control guarantees through e-values. Consider a setting where one has L outcomes, p_x treatments and p_z control covariates, with n observations for each. Said observations may be potentially split across $p_g \geq 1$ subpopulations. Let $y^{(l)} \in \mathbb{R}^n$ for $l = 1, \ldots, L$ be the observed outcomes, X the $n \times p_x$ matrix and Z the $n \times p_z$ with the treatment and control covariates (respectively), and G a $n \times p_g$ matrix enconding the subgroup information (if one is not interested in subgroups, then G is omitted). Let x_i, z_i and g_i be the ith row in X, Z and

G respectively. Throughout, we denote expectations by $\mathbb{E}$, probabilities by $\mathbb{P}$, and by $\mathbb{E}_P$ and $\mathbb{P}_P$ expectations and probabilities when data are generated by a specific distribution P. Further, we denote by p the density (or probability mass) function associated to a distribution P, and by q those associated to a distribution Q.

A generalized linear model for outcome l assumes that

$$F(\mathbb{E}(y_i^{(l)} \mid x_i, z_i, g_i)) = \beta_0^{(l)} + z_i^T \eta_z^{(l)} + g_i^T \eta_g^{(l)} + \sum_{j=1}^{p_x} (\beta_j^{(l)} + g_i^T \delta_j^{(l)}) x_{ij}, \tag{1}$$

where F is the link function (e.g., the identity for linear regression, or the logit for logistic regression), $\beta_0^{(l)} \in \mathbb{R}$ is the intercept, $\beta_j^{(l)} \in \mathbb{R}$ the average effect of treatment j on outcome l, $\delta_j^{(l)}$ captures subpopulation-specific effects, and $\eta_z^{(l)} \in \mathbb{R}^{p_z}$ and $\eta_g^{(l)} \in \mathbb{R}^{p_g}$ are coefficients associated to control covariates and subgroups respectively. We denote the whole parameter vector by $\theta^{(l)} = (\{\beta_j^{(l)}\}_{j=0}^{p_x}, \{\delta_j^{(l)}\}_{j=1}^{p_x}, \eta_z^{(l)}, \eta_g^{(l)})$. Importantly, to be able to interpret $\beta_j^{(l)}$ as an average treatment effect, $g_i \in \mathbb{R}^{p_g}$ must encode subpopulation information using a sum-to-zero constraint. Specifically, let $\rho_k = n_k/n$ where n_k is the number of observations coming from subpopulation k, we define

$$g_{ik} = \begin{cases} 1 - \rho_k, \text{ if observation } i \text{ comes from subpopulation } k \\ -\rho_k, \text{ otherwise} \end{cases}$$

for $k = 1, \ldots, p_g$, which implies that

$$\sum_{i=1}^{n} g_{ik} = n_k(1 - \rho_k) + (n - n_k)(-\rho_k) = n_k \frac{n - n_k}{n} - (n - n_k)\frac{n_k}{n} = 0.$$

We define the effect of treatment j on outcome l for individual i in subpopulation k as

$$\beta_{jk}^{(l)} = \frac{\partial}{\partial x_{ij}} F(\mathbb{E}(y_i^{(l)} \mid x_i, z_i, g_i)) = \beta_j^{(l)} + g_i^T \delta_j^{(l)} = \beta_j^{(l)} + (1 - \rho_k)\delta_{jk}^{(l)} - \sum_{m \neq k} \delta_{jm}^{(l)} \rho_m.$$

The corresponding average treatment effect (ATE) across all individuals is

$$\frac{1}{n} \sum_{i=1}^{n} \frac{\partial}{\partial x_{ij}} F(\mathbb{E}(y_i^{(l)} \mid x_i, z_i, g_i)) = \beta_j^{(l)} + \frac{1}{n} \left(\sum_{i=1}^{n} g_i^T \right) \delta_j^{(l)} = \beta_j^{(l)}.$$

We consider a multiverse analysis where one is interested in estimating the ATEs $\beta_j^{(l)}$ for $l = 1, \ldots, L$ and $j = 1, \ldots, p_x$, and performing the corresponding Lp_x hypothesis tests

$$H_{lj} : \beta_j^{(l)} = 0, \beta_{j'}^{(l)} \neq 0 \text{ for } j' \neq j$$

$$\overline{H}_l : \beta_{j'}^{(l)} \neq 0 \text{ for } j' \in \{1, \ldots, p_x\} \tag{2}$$

That is, the null hypothesis H_{lj} is that treatment j has zero effect on outcome l, without any restrictions on the remaining treatment effects. And there is a global alternative hypothesis $\overline{H}_l$, shared across all treatments, that all treatment effects are non-zero. We denote by $\mathcal{H}_{lj}$ the set of probability distributions such that H_{lj} holds, and by $\overline{\mathcal{H}}_l$ the set where none of the H_{lj}'s hold (that is, $\overline{\mathcal{H}}_l$ is the complement of $\bigcup_{j=1}^{p_x} \mathcal{H}_{lj}$). In the event that one is also interested in testing the effect within subpopulations, there are Lp_xp_g further tests corresponding to $H_{ljk} : \beta_{jk}^{(l)} = 0$ for $l = 1, \ldots, L$, $j = 1, \ldots, p$ and $k = 1, \ldots, p_g$. To ease notation we drop the subscript k in the exposition below, that is we describe the case where there are no subpopulations, but our framework remains readily applicable when sub-populations are also of interest.

Although not stated explicitly, none of the remaining parameters in $\theta^{(l)}$ are assumed to be zero, neither under the null nor the alternative. In particular, the control covariates effects η_z are always assumed to be non-zero. This is contrast to SCA [1] or its Bayesian counterpart [2], where one also considers covariate subsets. As discussed in Section 1, SCA computes a median over subsets of control covariates that has no statistical interpretation. Its Bayesian counterpart computes a weighted average where weights are given by posterior model probabilities, which as $n \to \infty$ converges to performing inference under a model that includes only the controls that have non-zero effects. While this implies an asymptotic gain in statistical power, one does not obtain strict type I error control for finite n. Hence our strategy here of including all controls, as an alternative that is more conservative but strictly controls false positives.

Given the potentially large number of multiple tests being conducted, it is important to assess statistical significance in a way such that some measure of false positives is controlled. Specifically, we consider the FDR [10]. Let $\gamma_{lj} = \mathrm{I}(\beta_j^{(l)} \neq 0)$ be the indicator that the null hypothesis H_{lj} does not hold, let $\hat{\gamma}_{lj} = 1$ if we reject H_{lj} and $\hat{\gamma}_{lj} = 0$ otherwise. Then the number of false positive is $\sum_{l,j} \gamma_{lj}\hat{\gamma}_{lj}$, the number of rejected null hypotheses is $\sum_{l,j} \hat{\gamma}_{lj}$, and $\mathrm{FDR} = \mathbb{E}(\mathrm{FDP})$, where

$$\mathrm{FDP} = \begin{cases} \frac{\sum_{l,j} \gamma_{lj}\hat{\gamma}_{lj}}{\sum_{l,j} \hat{\gamma}_{lj}}, \text{ if } \sum_{l,j} \hat{\gamma}_{lj} > 0 \\ 0, \text{ otherwise} \end{cases}$$

and the expectation is with respect to the data-generating distribution. That is, the FDR is the expected false discovery proportion (FDP) and, if no hypotheses are rejected, then FDP is defined to be 0. If one is interested in testing subpopulation effects $\beta_{jk}^{(l)}$, then the FDR is analogously defined by taking sums over k as well.

3 e-values

E-values provide a framework to assess statistical significance that allows controlling the FDR, regardless of the dependence structure between the tests being performed. Formally, an e-value is the realization of a random variable called an e-variable, whose defining property is that its expectation under the null hypothesis is ≤ 1.

Definition 1 Consider a null hypothesis H_{lj} specifying a set of probability distributions $\mathcal{H}_{lj}$. A test statistic E_{lj} is an e-variable if and only if $\mathbb{E}_P(E_{lj}) \leq 1$ for any $P \in \mathcal{H}_{lj}$.

Larger values of E_{lj} provide stronger evidence for the alternative hypothesis. By Markov's inequality, if one rejects H_{lj} when $E_{lj} \geq 1/\alpha$, then the type I error is $\leq \alpha$:

$$\mathbb{P}_P\left(E_{lj} \geq \frac{1}{\alpha}\right) \leq \alpha \mathbb{E}_P(E_{lj}) \leq \alpha,$$

for any $P \in \mathcal{H}_{lj}$.

In Section 3.1 we review some strategies to define e-variables, and Section 3.2 reviews e-value-based procedures for FDR control.

3.1 Defining e-variables

There are many ways to define valid e-variables. While the Markov inequality guarantees type I error control for all e-variables, their statistical power may vary greatly. Efforts have been made to define e-variables that accumulate evidence against the null as fast as possible. Such optimality is usually defined in terms of maximizing $\mathbb{E}_Q[\log E_{lj}]$, where $Q \in \overline{\mathcal{H}}_l$ is a distribution within the alternative hypothesis. One possible definition of optimality is the so-called numeraire ([9], Chapter 6), an e-variable E^*_{lj} that is defined by E_{lj}/E^*_{lj} having expectation ≤ 1 under the alternative, where E_{lj} is any other e-variable. Another possible definition by [7] uses a worst-case optimality criterion called GROW. When testing a point null hypothesis versus a point null alternative, both definitions return the likelihood-ratio test as the optimal e-variable. This is of course not applicable to our setting, where the null hypothesis $\mathcal{H}_{lj}$ contains infinitely many distributions indexed by $\theta^{(l)}$, and so does the alternative $\overline{\mathcal{H}}_l$. In such more general settings, obtaining either the numeraire or the GROW-optimal e-variable is related to so-called reverse information projections (RIP), and it is a highly non-trivial task.

We review next some practical alternatives applicable to our generalized linear model framework (1): universal mixture e-variables, soft-rank e-variables and p-to-e calibration. Universal inference is a framework introduced by [17] where one splits the observed data $y^{(l)} = (y_0^{(l)}, y_1^{(l)})$, and $y_0^{(l)}$ is assumed independent from $y_1^{(l)}$. One then obtains the maximum likelihood estimate (MLE) for the density function under the null $\hat{p}_{lj} \in \mathcal{H}_{lj}$ using only $y_0^{(l)}$, that under the alternative $\hat{q}_l$ using only $y_1^{(l)}$, and the split likelihood-ratio e-variable proposed in [19] is defined as $E_{lj} = \hat{q}_l(y_0^{(l)})/\hat{p}_{lj}(y_0^{(l)})$. Universal mixture e-variables are a variation discussed in [9] (Chapter 5.2) that avoids using arbitrary data splits (as well as assuming independence) by considering a hybrid of frequentist and Bayesian hypothesis tests. Specifically, let $\hat{p}_{lj} \in \mathcal{H}_{lj}$ be the MLE under the null, now using the full data $y^{(l)}$. Define

$$E^{um}_{lj} = \frac{q_l(y^{(l)})}{\hat{p}_{lj}(y^{(l)})}, \tag{3}$$

where

$$q_l(y) = \int p(y \mid \theta^{(l)}) p(\theta^{(l)} \mid \overline{H}_l) d\theta_l$$

is the marginal (or integrated) likelihood with respect to a prior density $p(\theta_j^{(l)} \mid \overline{H}_l)$ that places full support on the alternative (i.e., such that the prior probability that $\theta_j^{(l)} \neq 0$ is 1). It is easy to show that E_{lj} is an e-variable for any choice of $p(\theta_j^{(l)} \mid \overline{H}_l)$. Since $\hat{p}_{lj}$ is the MLE under $\mathcal{H}_{jl}$, by definition we have that $p(y^{(l)}) \leq \hat{p}_{lj}(y^{(l)})$ for any $P \in \mathcal{H}_{lj}$, therefore

$$\mathbb{E}_P\left(E_{lj}\right) = \int \frac{q_l(y^{(l)})}{\hat{p}_{lj}(y^{(l)})} p(y^{(l)}) dy^{(l)} \leq \int \frac{q_l(y^{(l)})}{p(y^{(l)})} p(y^{(l)}) dy^{(l)} = \int q_l(y^{(l)}) dy^{(l)} = 1.$$

In our framework (1), to avoid subjectivity in the choice of $p(\theta_j^{(l)})$, one may consider Zellner's unit information prior [20], a classical default that has tight connections to the Bayesian information criterion [21]. Specifically, let $W = (X, Z, G)$ be the design matrix associated to the regression equation in (1) (if subpopulations are of interest, then the interaction terms GX are also included in W). Zellner's unit information prior is

$$\theta_j^{(l)} \sim N(0, n\sigma_l^2 (W^T W)^{-1}), \tag{4}$$

where $\sigma_l^2 > 0$ is the GLM dispersion parameter associated to $y^{(l)}$. This choice assumes that $W^T W$ is invertible, as is the case in our examples here, otherwise an alternative default is to replace $W^T W$ by $\mathrm{diag}(W^T W)$. Expression (3) requires obtaining an MLE for the Lp_x alternative hypotheses, which can be done efficiently and in parallel using any suitable optimization algorithm. It also requires obtaining L marginal likelihoods $q_l(y)$, one for each outcome. In the linear regression case where $y^{(l)} \sim N(W\theta^{(l)}, \sigma_l^2 I)$, q_l has a closed-form. Specifically, if one sets an inverse Gamma prior $\sigma_l^2 \sim \mathrm{IG}(a/2, b/2)$, then

$$\frac{q_l(y^{(l)})}{\hat{p}_{lj}(y^{(l)})} = \frac{b^{a/2}\Gamma(\frac{n+a}{2})}{(n+1)^{d/2}\Gamma(a/2)} \frac{(2e/\hat{\sigma}_l^2)^{\frac{n}{2}}}{\left(l + \|y^{(l)}\|^2 - \frac{n}{n+1}(\hat{\theta}^{(l)})^T W^T W \hat{\theta}^{(l)}\right)^{\frac{n+a}{2}}}$$

where $\|\cdot\|^2$ is the squared Euclidean norm, $d = \dim(\theta^{(l)})$, $\hat{\theta}^{(l)} = (W^T W)^{-1} W^T y^{(l)}$ is the least-squares estimator and $\hat{\sigma}_l^2 = n^{-1}(\|y^{(l)}\|^2 - \|W\hat{\theta}^{(l)}\|^2)$ the MLE for the error variance. Outside of linear regression numerical integration methods must be used and, as long as the number of outcomes L is not too large, this can also be done quickly (and in parallel). One may approximate q_l using Laplace approximations, which is very fast as it requires a single optimization exercise for each outcome, and is very accurate as n grows ([22], [23] Lemma S3).

A second possible strategy is to use soft-rank e-variables [18]. These are typically based on permutation exercises akin to those used to obtain permutation-based p-values, therefore they are more computationally-intensive but as a positive aspect they are more reliable when our statistical model in (1) is misspecified (e.g., covariate effects are truly non-linear, errors are heteroskedastic, etc.). Let $T_0 \geq 0$ be a test statistic for the null hypothesis of interest, and let $T_1, \ldots, T_B$ be the realization of B test statistics that are exchangeable with T_0 under the null. The soft-rank e-variable is

$$E^{sr} = \frac{(B+1)T_0}{\sum_{b=0}^{B} T_b}. \tag{5}$$

In generalized linear models, T_0 can be the likelihood ratio associated to (2),

$$T_0 = \frac{\hat{q}_l(y^{(l)})}{\hat{p}_{lj}(y^{(l)})} \tag{6}$$

where recall that $\hat{p}_{lj}$ is the MLE for the density function under the null hypothesis H_{lj}, and $\hat{q}_l$ that under the alternative. The exchangeable likelihood ratio test statistics $T_1, \ldots, T_B$ can be obtained using permutations. Let $y^{(l,b)}$ be a permuted version of $y^{(l)}$ such that the null hypothesis H_{lj} holds, and define

$$T_b = \frac{\hat{q}_l^b(y^{(l,b)})}{\hat{p}_{lj}^b(y^{(l,b)})}$$

where $\hat{p}_{lj}^b$ are the MLEs under the null and alternative hypotheses respectively, computed on the permuted dataset b. We remark that although we denote permutations by $y^{(l,b)}$ for simplicity, permutations actually typically affect covariate values rather than the outcome. For example, in our simulations, permutation are obtained by first regressing the treatment of interest on all other covariates, permuting the residuals from such regression, and defining a synthetic treatment given by the expected value given other covariates plus a permuted residual. In such a permutation exercise, the outcome is conditionally independent from the treatment given other covariates, and therefore T_0 is exchangeable with $T_1, \ldots, T_B$ when H_{lj} holds.

The third and final option that we discuss here to obtain an e-variable is to use so-called p-to-e calibrators [9], which convert p-values into e-values. Let π_{lj} be a p-value for the null hypothesis H_{lj} in (2), two popular calibrators are

$$E^{c1}(\pi_{lj}) = \int_0^1 \kappa \pi_{lj}^{\kappa-1} \, \mathrm{d}\kappa = \frac{1 - \pi_{lj} + \pi_{lj} \log \pi_{lj}}{\pi_{lj} (\log \pi_{lj})^2}, \tag{7}$$

which uses a mixture over $\kappa \in (0, 1)$ of the calibrator function proposed in [24], and

$$E^{c2}(\pi_{lj}) = \pi_{lj}^{-1/2} - 1. \tag{8}$$

We refer to (7) as the mixture calibrator and to (8) as the inverse root calibrator. The family of p-to-e calibrators is much richer, with no single calibrator shown to uniformly outperform all others. See [9] (Chapter 2) for a review. In our setting one may take π_{lj} to be the likelihood ratio test p-value for (2). A theoretical caveat is that, outside of the Gaussian family, most implementations of the test rely on the chi-square asymptotic distribution of the likelihood ratio test. Since such a p-value is not exact for finite n, neither is the resulting calibrated e-value. However, in our simulations such e-values controlled the type I error adequately, already for moderately large n.

3.2 FDR control with e-variables

In this section we first review strategies to control the FDR with e-variables, and next compare them with the Benjamini-Yekutieli (BY) P-value adjustment procedure [11], which also controls FDR under arbitrary dependences between the tests. For e-variables, a simple strategy to control the FDR below any user-specified α is the e-BH procedure of [5]. An alternative is to use the closed e-BH procedure ([9], Chapter 9.6), denoted by $\overline{\text{e-BH}}$, which strictly dominates e-BH in terms of power at the cost of the method being slighly more involved. The e-BH procedure proceeds as follows. Let e_{lj} be the e-value, that is the observed value of the e-variable E_{lj}, for $l = 1, \ldots, L$ and $j = 1, \ldots, p_x$, and let $K = Lp_x$ be the number of hypotheses being tested. Let $e_{[k]}$ be the kth order statistic of e_{lj} across all (l, j), sorted from largest to smallest. The e-BH procedure at level α rejects all hypotheses with the k^* largest e-values, where

$$k^* = \max\left\{k \in \{1, \ldots, Lp_x\} : \frac{ke_{[k]}}{Lp_x} \geq \frac{1}{\alpha}\right\},$$

with the convention that $\max \emptyset = 0$. That is, the kth test effectively uses $ke_{[k]}/K$ as its e-value, and if we reject test k then we also reject all the previous tests. [5] proved that FDR $\leq \alpha K_0/K \leq \alpha$, where K_0 is the number of truly null hypotheses.

We now describe the closed e-BH procedure $\overline{\text{e-BH}}$. Let R_k be the set of indices corresponding to the k largest e-values, $\overline{\text{e-BH}}$ rejects all null hypotheses $k \leq \bar{k}$, where $\bar{k} = \max\{k \geq 0 : R_k \in C_\alpha\}$ and

$$C_\alpha = \left\{R : \frac{1}{|A|}\sum_{k \in A} E_k \geq \frac{\text{FDP}_A(R)}{\alpha}, \text{ for all } A \subseteq \{1, \ldots, K\}\right\}$$

where $\text{FDP}_A(R) = |A \cap R|/\max\{|R|, 1\}$ is the false discovery proportion associated to rejecting the hypotheses in R, if A were the subset of truly null hypotheses. In words, $R_{\bar{k}}$ is such that the average of the e-values over any other subset A exceeds $\text{FDP}_A(R)/\alpha$. $\overline{\text{e-BH}}$ controls the FDR at level α, and rejects at least as many hypothesis as the refined e-BH procedure. A disadvantage however is that finding $R_{\bar{k}}$ requires order K^3 operations, relative to e-BH which only requires K operations.

We now compare these procedures with BY, which is based on P-values rather than e-variables. If one is solely interested in FDR control, whether e-variable procedures are preferable to BY or not depends entirely on their respective statistical powers. Current

evidence suggests that no approach uniformly dominates the other. For example, [5] showed that e-BH can have higher power than BY in certain settings and lower on others, depending on how the e-variables are obtained and on their distribution. To gain some intuition, denote by K the total number of tests, by $P_{[k]}$ the kth ordered P-value and by $E_{[k]}$ the kth ordered e-variable. Recall that BY uses the threshold

$$P_{[k]} \leq \frac{\alpha k}{K c(K)}$$

where $c(K) = \sum_{i=1}^{K} 1/i \approx \ln(K) + 0.577$ is the Kth harmonic number. In contrast, e-BH sets the threshold

$$\frac{1}{E_{[k]}} \leq \frac{\alpha k}{K}.$$

There are therefore two issues driving how these two thresholds behave. First, as pointed out by [5], e-BH avoids the $\ln(K)$ term in the denominator, hence the threshold scales better as K grows. Second, for any fixed K the relative powers depend on the magnitude of $1/E_{[k]}$ versus that of $P_{[k]}$. For example, for the mixture calibrator e-variable (which has a fairly competitive power in our experiments), we have that

$$E_{[k]} = \frac{1 - P_{[k]} + P_{[k]} \ln(P_{[k]})}{P_{[k]} (\ln P_{[k]})^2},$$

where the right-hand side converges to $1/[P_{[k]} (\ln P_{[k]})^2$ as $P_{[k]} \to 0$. Hence and roughly speaking, for small $P_{[k]}$, applying e-BH to the mixture calibrator implies a threshold

$$P_{[k]} (\ln P_{[k]})^2 \leq \frac{\alpha k}{K}.$$

Therefore, if $(\ln P_{[k]})^2 > \ln(K) + 0.577$ then one expects BY to be less conservative than e-BH, and otherwise e-BH should be less conservative. Note however that, because of the step-up nature of e-BH and BY, the hypotheses that are actually rejected are driven by the largest e-variable and smallest P-value (respectively) satisfying the threshold. That is, the power of these procedures is not by the most extreme e-variables / P-values. Recall also e-BH is not the most powerful FDR controlling procedure for e-variables, for the closed e-BH procedure discussed above has higher power. The reason for discussing e-BH here is that it leads to more tractable expressions that provide some insights.

Another consideration is that, beyond power comparisons, there are situations where e-variables can have appealing properties. One such property that can be relevant for multiverse analysis is evidence aggregation over various hypotheses. Suppose that one considers a set A of related hypotheses (such as the effect of a treatment on multiple related outcomes), one can obtain an overall e-variable E_A by aggregating the e-variables from the individual hypotheses. A simple such choice is defining the overall e-variable as the average (or weighted average) of the e-variables for the individual

hypotheses. If all the hypotheses in A are true, then $\mathbb{E}[E_A] = \sum_{k\in A} \mathbb{E}[E_{[k]}]/|A| \leq 1$. This can be used to test the intersection null hypothesis, that is that all null hypotheses in A are true.

Another property, albeit less relevant to standard multiverse applications, is the anytime validity of e-variables. This is useful in sequential settings where one decides to continue or stop experimentation based on the data that was observed so far, in a manner that was not pre-defined in advance. In such settings, P-values cannot be calibrated to control type I error, whereas e-variables continue to be valid, provided that they are constructed to satisfy certain martingale properties. Although we are not aware of applications that have made such uses for multiverse analysis, one could consider a setting where a multiverse analysis is conducted using an available dataset, and based on those findings one decides to gather additional data to gain more information on some of the specific treatments or outcomes. Assuming the additional data to be independent from the previous data, one easily define e-variables that combine the evidence across both datasets, for example by taking the products of the e-variables obtained in each set. As a word of caution however, a repeated application of e-variable multiplicity adjustments does not guarantee FDR control. For example, applying e-BH once all data has been obtained controls the FDR, but applying e-BH for the first dataset and for the combined dataset does not control the FDR.

4 Simulation study

Given that there are several strategies to define e-variables (Section 3.1), we first conduct in Section 4.1 a simulation exercise to compare their statistical powers, in the context of single-outcome linear and logistic regressions, with multiple treatments and control covariates. Subsequently, in Section 4.2 we extend the simulations to a multiverse setting where one considers several outcomes, focusing on the two most powerful e-values found in Section 4.1. We also compare their power to that of BY P-value adjustment.

4.1 Comparing e-variable methods

Table 1 Proportion of type I errors for $E \in \{E^{um}, E^{sr}, E^{c1}, E^{c2}\}$ and significance levels $\alpha \in \{0.01, 0.05\}$ in logistic regression

n	E^{um}		E^{sr}		E^{c1}		E^{c2}	
	$\alpha = 1\%$	$\alpha = 5\%$	$\alpha = 1\%$	$\alpha = 5\%$	$\alpha = 1\%$	$\alpha = 5\%$	$\alpha = 1\%$	$\alpha = 5\%$
100	0	0	0	0	0	0.01	0	0.01
500	0	0	0	0	0	0.01	0	0.01
1500	0	0	0	0	0	0.01	0	0.01
2000	0	0	0	0	0	0	0	0

We assess the type I error control and power of the four strategies to obtain e-values discussed in Section 3.1: universal mixture and soft-rank e-values and the mixture and inverse root calibrators in (7)-(8). We run a simulation study in Gaussian linear

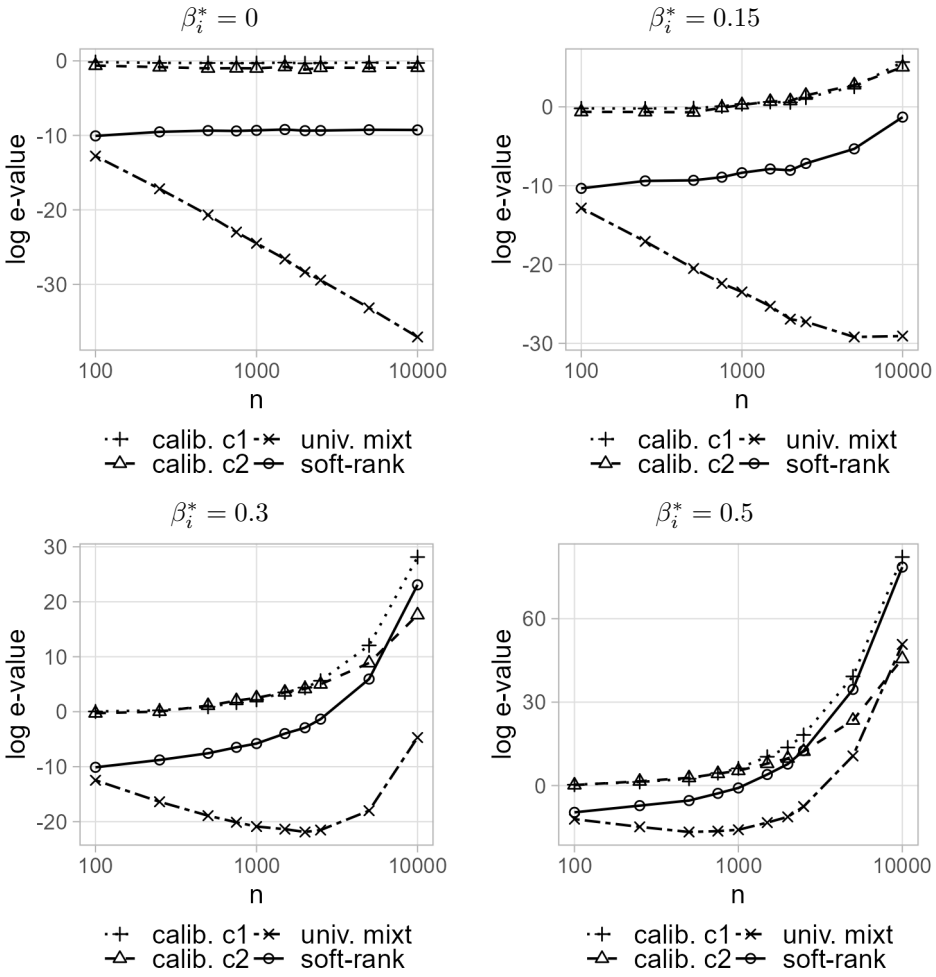


Fig. 1 Average log e-values in logistic regression for $\beta_i^* = 0$ (top left), $\beta_i^* = 0.15$ (top right), $\beta_i^* = 0.3$ (bottom left) and $\beta_i^* = 0.5$ (bottom right) for universal mixture and soft-rank e-values, mixture (calib.c1) and inverse root p-to-e calibrators (calib.c2)

regression and in logistic regression. We considered sample sizes n ranging from 100 to 10,000 and 10 covariates. The covariates were generated from a zero-mean multivariate Gaussian with pairwise covariance 0.5. Covariates were scaled to unit sample variance. For each family and each sample size, we simulate 100 datasets with data-generating regression coefficients $\beta_i^* \in [0, 1]$, $i = 1, \ldots, 10$.

Table 1 reports the proportion of type I errors over the 100 repetitions for selected sample sizes $n \in \{100, 500, 1500, 2000\}$ in logistic regression. All considered e-variables control the type I error below the specified significance level α, in fact they behaved conservatively in that the proportions of type I errors were smaller than the desired level. Figure 1 plots the average log e-values over the 100 repetitions for $\beta_i^* \in \{0, 0.15, 0.3, 0.5\}$ in logistic regression. Recall that larger e-values indicate

stronger evidence against the null hypothesis. For data-generating $\beta_i^* \neq 0$ and small n, the calibrated e-values E^{c1} and E^{c2} clearly dominate the universal mixture and soft-rank e-variables E^{um} and E^{sr}, accumulating stronger evidence in favor of the alternative. For large n, the calibrated e-variable E^{c2} became less competitive whereas the soft-rank e-variable E^{sr} became more competitive, nearly matching the mixture calibrated e-variable E^{c1} in some scenarios.

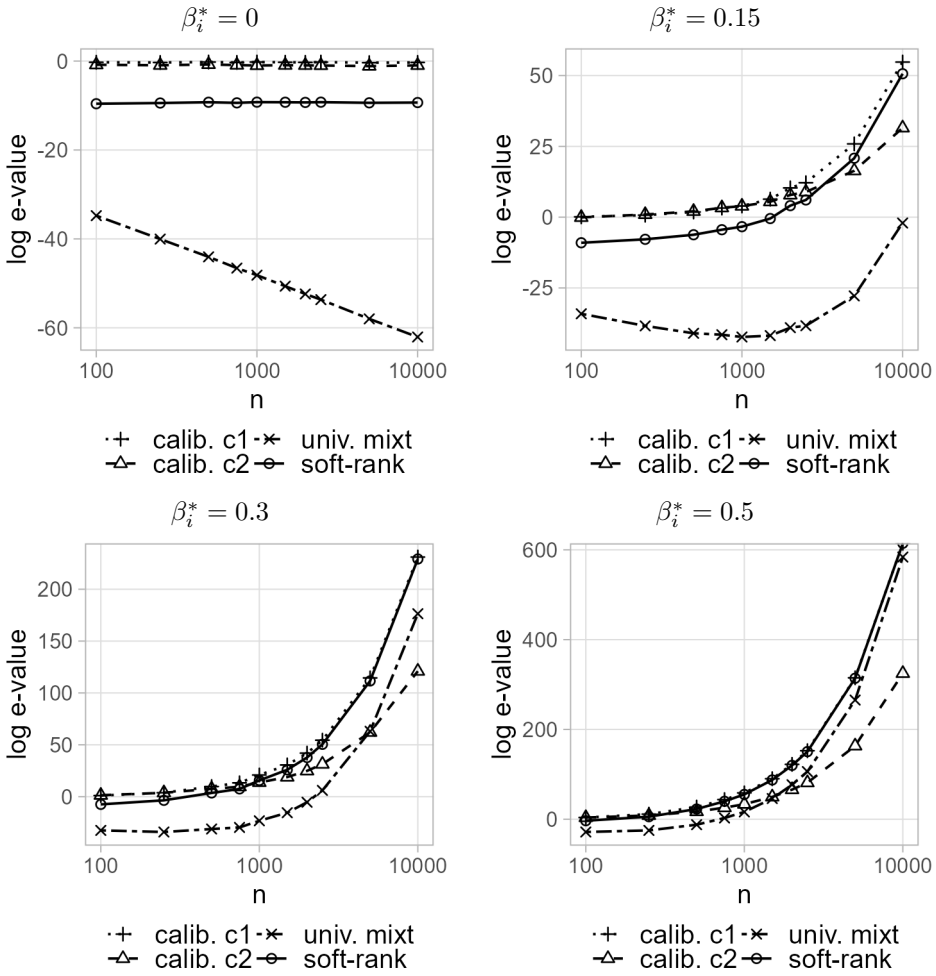


Fig. 2 Average log e-values in linear regression for $\beta_i^* = 0$ (top left), $\beta_i^* = 0.15$ (top right), $\beta_i^* = 0.3$ (bottom left) and $\beta_i^* = 0.5$ (bottom right) for universal mixture and soft-rank e-values, mixture (calib.c1) and inverse root p-to-e calibrators (calib.c2)

Figure 2 reports the average log e-value for $\beta_i^* \in \{0, 0.15, 0.3, 0.5\}$ over the 100 repetitions in Gaussian linear regression. The findings are similar to those from logistic regression. Further, there were no type I errors in our linear regression simulations

for selected samples sizes $n \in \{100, 500, 1500, 2000\}$ for $\alpha \in \{0.01, 0.05\}$. In summary, out of the assessed e-variables, we recommend the mixture calibrator E^{c1} to test the null hypotheses in (2), as it accrues stronger evidence for truly alternative hypotheses while appropriately controlling type I error. We remark that such e-values are only properly calibrated when the assumed model is correct (or approximately so), otherwise permutation-based soft-rank e-variables may be preferable for being more robust to model misspecification.

4.2 Multiverse analysis simulation

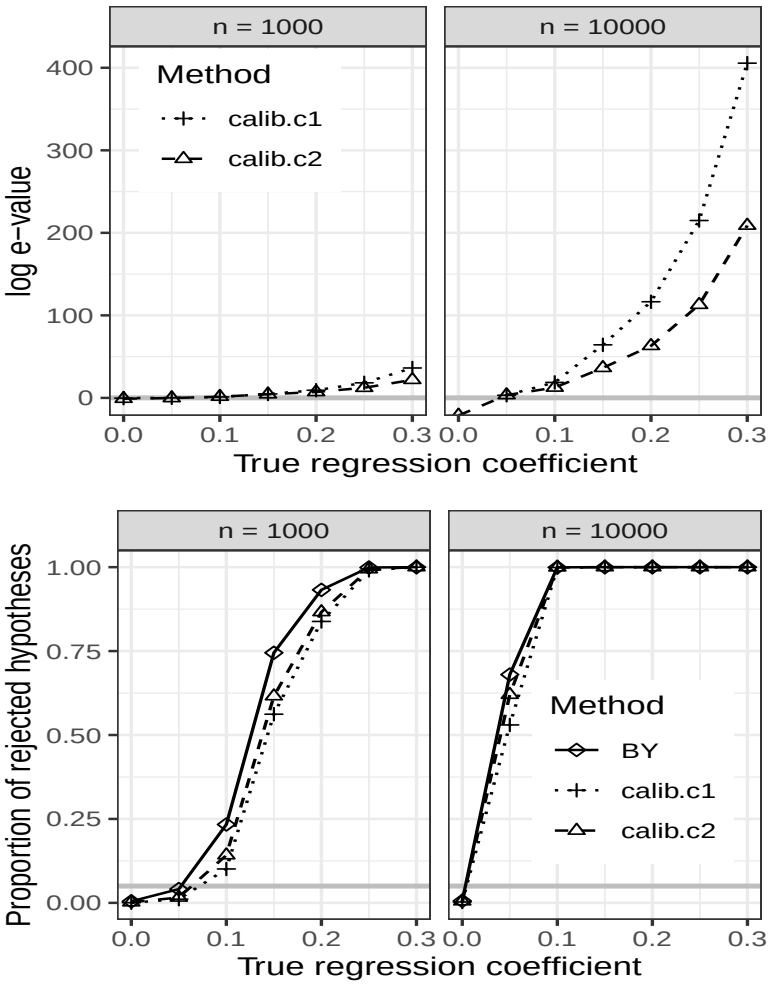


Fig. 3 Multiverse simulation. Top: average log e-values for mixture (calib.c1) and inverse root p-to-e calibrators (calib.c2). Bottom: proportion of rejected hypothesis vs. data-generating β_i^*.

We considered a multiverse analysis that mimics the teenager application in Section 5, with $p_x = 5$ treatments and $p_z = 14$ controls, whose values were generated drawing with replacement observations from the teenager study. We set the sample size to $n = 1,000$ and $n = 10,000$. We considered a moderately-sized multiverse featuring $L = 5$ outcomes and a larger multiverse with $L = 20$ outcomes, leading to a total of 20 and 80 hypothesis being tested (respectively).

We simulated the outcomes from linear regression models with independent standard Gaussian errors. The control covariate coefficients were $\eta_z^{(l)} = (1, 0.9, 0.8, \ldots, 0.1, 0, 0, 0, 0)$ for all outcomes $l = 1, \ldots, L$. In the multiverse with $L = 5$ outcomes, we set the treatment effects to $\beta^{(1)} = (0, 0, 0, 0, 0)$, $\beta^{(2)} = (0, 0, 0.05, 0.1, 0.15)$, $\beta^{(3)} = (0, 0, 0.1, 0.15, 0.2)$, $\beta^{(4)} = (0, 0, 0.15, 0.2, 0.25)$, $\beta^{(5)} = (0, 0, 0.2, 0.25, 0.3)$. That is, all treatment effects are 0 for outcome 1, and for the remaining treatments two are 0 and three are non-zero. In the multiverse with $L = 20$ outcomes, we used the same $(\beta^{(1)}, \ldots, \beta^{(5)})$ for outcomes 1-5, and again the same coefficients for outcomes 6-10, 11-15, and 16-20.

We only considered the two e-variables that exhibited the highest power in Section 4.1, namely the mixture E^{c1} and inverse root E^{c2} calibrated e-variables, as well as BY-adjusted P-values.

For each n and L combination we simulated 1,000 independent datasets, and we report averaged results in Figures 3 and **??** for $L = 5$ and $L = 20$ respectively. Similar to our findings from Section 4.2, E^{c1} exhibited a larger mean log-evidence than E^{c2} (top panel) when truly $\beta_i^{(l)} \neq 0$, and similar mean when truly $\beta_i^{(l)} = 0$. Interestingly, their statistical power when using the closed e-BH procedure to control the FDR below 0.05 was very similar (bottom panel). That is, while the mean log E^{c1} tended to be higher than the mean log E^{c2}, the difference was mostly observed in situations where both were fairly high, and therefore had a negligible effects on the statistical power at a 0.05 FDR cutoff. Figure 3 also shows that the power for BY-adjusted P-values was slightly higher than for either e-variable. If one is purely interested in FDR control, in our considered settings there are no benefits in using e-variables over BY-adjusted P-values. We remark however that e-variables may still be interesting for easily combining evidence across hypotheses, and across independent datasets, as illustrated in Section 5.2.

5 Teenager mental well-being and technology use

We apply our multiverse framework to study the association between technology and teenager mental well-being in the 2015 survey of the Millennium Cohort Study (MCS) in Section 5.1. The MCS is a socio-economic and health survey tracking a representative sample of children born in the UK around year 2000 throughout their life. The sample size is $n = 11,884$ teenagers, with $> 8,000$ complete observations (Table 2), and features several outcome measures of mental well-being, $p_x = 5$ treatments measuring the use of various technologies, and $p_z = 14$ control covariates. Using Specification Curve Analysis, [3] concluded that the association between technology and teenager well-being was not practically relevant. Here, we estimate and obtain e-values

	Outcome reported	Complete observations
Depressed (adolescent)	11200	8351
Low self-esteem (adolescent)	11165	8311
High total difficulties (parent)	11471	8238
High emotional problems (parent)	11486	8240
High conduct problems (parent)	11488	8243
High hyperactivity/inattention (parent)	11481	8243
High peer problems (parent)	11491	8244
Low pro-sociality (parent)	11489	8243

Table 2 Number of observations and of complete observations (no missings in the treatments nor in the controls) for each outcome in the MCS dataset

Table 3 MCS Data. Rejected hypothesis by the refined e-BH procedure using mixture calibrated e-variables E^{c1} (top 4 rows) or by BY P-value adjustment (top 7 rows) with FDR level 0.05, estimated odds ratio and 95% e-confidence intervals

Outcome	Treatment	Odds ratio	95% conf. int.	e-value	BY P-value
Low self-esteem	internet	4.20	(1.72,10.39)	37,790	1.75×10^{-5}
Depressive symptoms	social media	3.02	(1.37,6.68)	1,851	2.92×10^{-4}
High peer problems	internet	2.14	(1.21,3.79)	757	5.68×10^{-4}
Depressive symptoms	internet	2.81	(1.27,6.32)	396	9.38×10^{-4}
Low pro-sociality	electronic games	2.08	(1.07,4.07)	52.6	0.001
Total difficulties	electronic games	1.85	(1.04,3.31)	36.9	0.013
Low self-esteem	social media	2.34	(0.97,5.68)	15.7	0.036

for the association between each (outcome, treatment) combination and we use the $\overline{\text{e-BH}}$ procedure to control the FDR, obtaining opposite conclusions to [3].

In Section 5.2 we illustrate how to use e-values to easily combine evidence across hypotheses and across datasets. First, we obtain the overall evidence that each treatment is associated with any outcome, based on the MCS data. Second, we combine the evidence from the MCS study with that provided by the independent Monitoring the Future (MtF) study [25], which was analyzed by [26].

5.1 Multiple treatment-outcome associations

We consider $L = 8$ outcomes: self-assessed depressive symptoms (Mood and Feelings Questionnaire) and self-esteem (Rosenberg scale), as well as parent-assessed total difficulties, emotional problems, conduct problems, hyperactivity/inattention, peer problems and pro-sociality (all Strengths and Difficulties Questionnaire). There are $p_x = 5$ treatments: time spent watching TV, playing video games, using the internet, using social media, and owning a personal computer. In all regressions, we include $p_z = 14$ control covariates: gender, age, BMI, (self-reported) educational motivation, mother's ethnicity, (self-reported) closeness to parents, presence of natural father, time spent with primary caretaker (PC), PC's word activity score, PC's employment

status, own longstanding illness, PC's psychological distress, number of siblings and household income.

The data were prepared as described in [2]. In particular, all treatment variables but TV usage were treated as continuous variables, normalized such that 0 means "no usage" and 1 means maximum reportable usage. TV usage was discretized into low, medium and high usage because its association with most outcomes was not linear (although it was monotone). The outcome variables are binary and were obtained by binning originally continuous data with validated cutoffs for abnormal behavior. See [2] (Section S4) for further details on data pre-processing. Table 2 reports the number of individuals reporting the outcomes and that of complete observations, with no missings on the treatments nor in the controls. These missings were mostly due to control covariates, with 3,446 individuals not reporting at least one control, and only 376 failing to report a treatment. Our analysis is based on these complete observations, and therefore our findings should be interpreted as being conditional on this population. We remark that one could consider several strategies to handle the missing data and view them as a data analytic choice within the multiverse analysis. While interesting, such comparisons are beyond are scope.

As an exploratory analysis, we plot in Figure 4 the mean treatment effects when fitting a logistic regression on each outcome against treatment and control variables. The reported 95% confidence interval are e-confidence intervals based on e-variable E^{c1}, that is, the ranges of treatment effect values for which $E^{c1} < 1/0.05$ [9]. In other words, an e-confidence interval captures the ranges of null hypothesis that are not rejected for a given level, similarly to standard confidence interval derived from test statistics. A total of 7 treatments appear significant based on e-confidence intervals. A more liberal approach considering standard likelihood ratio p-values individually and a 5% cutoff would lead to 12 significant treatments. These e-intervals and p-values do not however account for the multiplicity and correlation of the hypotheses being tested.

Following our proposed framework, we compute next e-values for each outcome-treatment pair and apply the closed $\overline{\text{e-BH}}$ procedure (Section 3.2). In particular, we used the mixture calibrator e-variable E^{c1} in (7), as it exhibited the largest power in our simulations. Before applying the e-BH procedure, there were six hypotheses for which $1/E^{c1} > 20$, a thresholding which controls the individual type I error for each hypothesis at the 0.05 level. Table 3 reports the subset of these that were rejected by the refined e-BH procedure, when controlling the FDR below 0.05. Using the inverse root e-variable lead to the same conclusions albeit with smaller e-values. Alternative, less powerful, choices of e-variables in Section 3.1 (universal mixture, soft-rank) lead to no rejection in this dataset after the refined e-BH adjustment. Applying $\overline{\text{e-BH}}$ instead of the refined e-BH procedure did not alter any of the results. In total, four hypotheses are rejected by our framework suggesting a significant effect of technology on teenager well-being contrary to the original study in [3]. Specifically, a strong internet usage was found to be significantly associated with low self-esteem and depressive symptoms as self-assessed by the teenagers, and high peer problems as assessed by their parents. High social media usage was also found to be significantly associated with self-assessed depressive symptoms. We remark that the odds-ratio for these associations were in

the range $[2, 4]$, that is these findings are not only statistically significant but also practically relevant.

The main reason why our findings differ from the original study in [3] is that the latter used Specification Curve Analysis to estimate a median treatment effect across all outcomes and treatments (as well as also across all possible control covariate configurations). As shown in Figure 4 there is significant heterogeneity of the effects across outcomes and treatments. For example, the point estimate for the association between high TV usage and most outcomes was negative (suggesting that TV usage might be associated with lower chances of poor mental well-being), whereas for social media and internet use the estimates were positive for self-assessed outcomes like depression and low self-steem and negative for parent-assessed outcomes like high peer problems and low pro-sociality. The median of these heterogeneous effects is close to 0, which led [3] to conclude that the association between mental well-being and technology use was too small to warrant policy change. We argue that in the presence of strong treatment effect heterogeneity, summaries such as the median can mask the presence of important individual treatment effects. We remark that there are observational studies [27], randomized controlled trials [28, 29] and natural experiments [30] indicating that there is a negative effect of social media on self-reported mental well-being, aligning with our findings.

5.2 Combining evidence

Table 4 Combining e-values. Evidence against the global null hypothesis that each treatment has no effect on any outcome in the MCS data, and combining evidence on treatment effects on depression between the MCS and MtF studies (average calibrated e-variable E^{c1})

Dataset	Outcome	Treatment	evalue
MCS	All 8 MCS outcomes	Internet	4869.0
MCS	All 8 MCS outcomes	Social media	236.0
MCS	All 8 MCS outcomes	Electronic games	13.6
MCS	All 8 MCS outcomes	TV	1.17
MCS	All 8 MCS outcomes	Computer	1.02
MCS + MTF	Depressed (adolescent)	Social media	1.99×10^{16}
MCS + MTF	Depressed (adolescent)	TV	101,186

To illustrate that e-variables allow combining evidence across multiple tests in a simple and intuitive manner, we perform two analyses. The first analysis is reported in rows 1-5 of Table 4. For each treatment, it shows an e-variable that provides evidence against the global null hypothesis that the treatment has no effect on any of the outcomes in the MCS study. Specifically, said e-variable is obtained as the average of the e-variables across the 8 outcomes. Table 4 shows that there is strong evidence that internet and social media use have an effect, whereas for electronic games there is moderate evidence and for TV and computer usage there is very little evidence. We emphasize that one could also use P-values to test these global null hypotheses.

For example, Bonferroni-corrected P-values also control the type I error under the global null and under arbitrary dependences between the hypotheses. In the MCS data, the Bonferroni correction obtains statistically significant evidence (at the 0.05 level) for internet, social media and electronic games, providing a similar message to that in Table 4. Hence, in this example the main appeal for e-value aggregation is the interpretability of the provided overall measures of evidence, rather than on statistical power.

The second analysis is shown in the last 2 rows of Table 4. It illustrates that, after seeing the MCS data, one can proceed to obtain more data to obtain further evidence, and easily aggregate that evidence with the e-values in Table 3, while still controlling type I error for each individual hypotheses. Specifically, [26] studied the association between depression (outcome) and social media and TV (treatments) using the MtF data. Since MCS and MtF are independent studies conducted in different countries, we may obtain a combined e-value for this outcome and 2 treatments by taking the product of their e-values in the MCS study (Table 3) and that in the MtF study. Table 2 in [26] reports a significant association between depression and social media and TV (P-values 6.69×10^{-17} and 2.58×10^{-8} respectively), in an analysis that was adjusted by gender, race, socio-economic status, grade and region where they lived, and which included $n = 77,487$ individuals. The mixture calibrator gave e-values 1.08×10^{13} for social media and $126,827$ for TV. Taking the product with the e-values obtained in the MCS data (1,851 and 0.798) gives aggregated e-values 1.99×10^{16} and 101,186 for social media and TV. Note that MCS did not provide evidence for the association with TV and, after failing to reject that null hypothesis with a P-value, one cannot gather more data while controlling type I errors. In contrast, e-values control type I error in such sequential settings.

6 Discussion

Multiverse analysis is a useful tool to descriptively assess how the association between treatments and outcomes varies as one considers different treatments, outcomes, subpopulations of individuals, control covariates, or other analysis choices. When purely used in such a descriptive manner, it can help flag whether treatment-outcome associations are robust across analysis specifications or not. A more contentious issue is when a multiverse analysis is used for formal statistical inference. In such situations, one must be careful to assess properly defined statistical estimands and to properly account for possible treatment effect heterogeneity. In particular, we do not recommend reporting averaged treatment effect estimates across all analysis specifications, except for fairly rare circumstances where the treatment and specifications are effectively viewed as exchangeable. Instead, we recommend reporting treatment effects for all analysis specifications separately and accounting for the multiple comparisons being performed. This ensures statistical validity but has a cost in statistical power, which can be a practical issue when either the sample size of the treatment effect are relatively small. For example, in the teenager data a previous analysis had found that the average effect across all treatments and outcomes was practically irrelevant,

whereas assessing each treatment-outcome association separately reveals strong and very practically-relevant associations.

A further statistical issue is that many false positive control methods make strong assumptions about the hypotheses being performed (e.g., independence or positive dependence across hypotheses, many tests being performed, etc.). E-values may be an appealing FDR control strategy in this setting because they relax these assumptions, in particular they remain valid regardless of the dependence structure across the statistical tests and of the number of hypotheses tested. A concern however is that e-values can be conservative, in the sense of having low statistical power to detect treatment effects when they truly exist. Our simulations show that this issue is much more marked for some e-values than for others, and these findings were confirmed in the teenager mental well-being data. Mixture calibrated e-variables worked best in our experiments, whereas on the other end of the spectrum universal mixture e-variables were extremely conservative in our examples, to the point of being practically useless unless n or the treatment effects are extremely large. Permutation-based soft-rank e-variables were less powerful than mixture calibrated e-variables, particularly for small n, but they may be appealing in settings where the model is misspecified.

In conclusion, our results suggest that e-variables can be a useful complement to multiverse analyses, provided that one wishes to perform hypothesis tests. However, they also point to significant limitations in terms of statistical power. Hence, e-values should be viewed as a conservative procedure that is best suited for at least moderately large sample sizes and treatment effect sizes. We also remark that e-values are a recent framework and that there is active research in designing more powerful procedures, which could improve their properties when used within multiverse analyses.

Declarations

The authors thank Christoph Semken for sharing the data and scripts for the teenager well-being example.

Funding. DR was funded by grant 0045 within the ICREA Academia program from AGAUR (Generalitat de Catalunya), and by grant PID2022-138268NB-I00 by Ministerio de Ciencia e Innovación (MICIN)/Agencia Española de Investigación(AEI)/10.13039/501100011033 /Fondo Europeo de Desarrollo Regional (FEDER). Co-funded by the European Union (ERC, BigBayesUQ, project number: 101041064). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them.

Ethics approval. Not applicable.

Consent. Not applicable.

Data. The data is available on the dedicated Github repository.

Code availability. The code is available on the dedicated Github repository.

Author's contribution statements. All authors contributed equally to this work.

References

[1] Simonsohn, U., Simmons, J.P., Nelson, L.D.: Specification curve analysis. Nature human behaviour **4**(11), 1208–1214 (2020)

[2] Semken, C., Rossell, D.: Specification analysis for technology use and teenager well-being: Statistical validity and a bayesian proposal. Journal of the Royal Statistical Society C **71**(5), 1330–1355 (2022)

[3] Orben, A., Przybylski, A.K.: The association between adolescent well-being and digital technology use. Nature human behaviour **3**(2), 173–182 (2019)

[4] Kass, R.E., Raftery, A.E.: Bayes factors. Journal of the american statistical association **90**(430), 773–795 (1995)

[5] Wang, R., Ramdas, A.: False discovery rate control with e-values. Journal of the Royal Statistical Society B **84**(3), 822–852 (2022)

[6] Vovk, V., Wang, R.: E-values: Calibration, combination and applications. The Annals of Statistics **49**(3), 1736–1754 (2021)

[7] Grünwald, P., Heide, R., Koolen, W.: Safe testing. Journal of the Royal Statistical Society B **86**(5), 1091–1128 (2024)

[8] Chugg, B., Ramdas, A., Grünwald, P.: E-values as statistical evidence: A comparison to bayes factors, likelihoods, and p-values. arXiv **2603.24421**, 1–34 (2026)

[9] Ramdas, A., Wang, R.: Hypothesis testing with e-values. Foundations and Trends® in Statistics **1**(1-2), 1–390 (2025)

[10] Benjamini, Y., Hochberg, Y.: Controlling the false discovery rate: A practical and powerful approach to multiple testing. Journal of the Royal Statistical Society B **57**(1), 289–300 (1995)

[11] Benjamini, Y., Yekutieli, D.: The control of the false discovery rate in multiple testing under dependency. The Annals of Statistics **29**, 1165–1188 (2001)

[12] Storey, J.D.: The positive false discovery rate: a Bayesian interpretation and the q-value. The Annals of Statistics **31**, 2013–2035 (2003)

[13] Efron, B., Tibshirani, R., Storey, J.D., Tusher, V.: Empirical Bayes analysis of a microarray experiment. Journal of the Americal Statistical Association **96**, 1151–1160 (2001)

[14] Storey, J.D.: The optimal discovery procedure: A new approach to simultaneous significance testing. Journal of the Royal Statistical Society B **69**, 347–368 (2007)

[15] Guindani, M., Müller, P., Zhang, S.: A Bayesian discovery procedure. Journal of the Royal Statistical Society, Series B **71**, 905–925 (2009)

[16] Efron, B.: Large-scale Inference: Empirical Bayes Methods for Estimation, Testing, and Prediction vol. 1. Cambridge University Press, Cambridge, United Kingdom (2012)

[17] Wasserman, L., Ramdas, A., Balakrishnan, S.: Universal inference. Proceedings of the National Academy of Sciences **117**(29), 16880–16890 (2020)

[18] Koning, N.W.: Measuring evidence against exchangeability and group invariance with e-values. arXiV **2310.01153**, 1–45 (2023)

[19] Ramdas, A., Manole, T.: Randomized and Exchangeable Improvements of Markov's, Chebyshev's and Chernoff's Inequalities. Statistical Science **41**(1), 121–142 (2026) https://doi.org/10.1214/24-STS952

[20] Zellner, A.: On assessing prior distributions and bayesian regression analysis with g-prior distributions. In: Bayesian Inference and Decision Techniques: Essays in Honor of Bruno de Finetti. North-Holland/Elsevier, Amsterdam; New York (1986)

[21] Schwarz, G.: Estimating the dimension of a model. The Annals of Statistics **6**, 461–464 (1978)

[22] Kass, R.E., Tierney, L., Kadane, J.B.: The validity of posterior expansions based on Laplace's method. Bayesian and likelihood methods in statistics and econometrics **7**, 473–488 (1990)

[23] Rossell, D., Rubio, F.J.: Additive Bayesian variable selection under censoring and misspecification. Statistical Science **38**(1), 13–29 (2021)

[24] Shafer, G., Shen, A., Vereshchagin, N., Vovk, V.: Test Martingales, Bayes Factors and p-Values. Statistical Science **26**(1), 84–101 (2011) https://doi.org/10.1214/10-STS347

[25] Miech, R.A., Johnston, L.D., O'Malley, P.M., Bachman, J.G., Schulenberg, J.E., Patrick, M.E.: Monitoring the future national survey results on drug use, 1975–2024: Secondary school students. Monograph, Institute for Social Research, The University of Michigan, Ann Arbor, MI (2025). https://monitoringthefuture.org

[26] Twenge, J.M., Joiner, T.E., Rogers, M.L., Martin, G.N.: Increases in depressive symptoms, suicide-related outcomes, and suicide rates among US adolescents after 2010 and links to increased new media screen time. Clinical psychological science **6**(1), 3–17 (2018)

[27] Valkenburg, P.M., Meier, A., Beyens, I.: Social media use and its impact on

adolescent mental health: an umbrella review of the evidence. Current Opinion in Psychology **44**, 58–68 (2022)

[28] Allcott, H., Braghieri, L., Eichmeyer, S., Gentzkow, M.: The welfare effects of social media. American Economic Review **110**, 629–676 (2020)

[29] Allcott, H., Gentzkow, M., Song, L.: Digital addiction. American Economic Review **112**, 2424–2463 (2022)

[30] Braghieri, L., Levy, R., Makarin, A.: Social media and mental health. London: Centre for Economic Policy Research. CEPR Discussion Paper **17252** (2022)

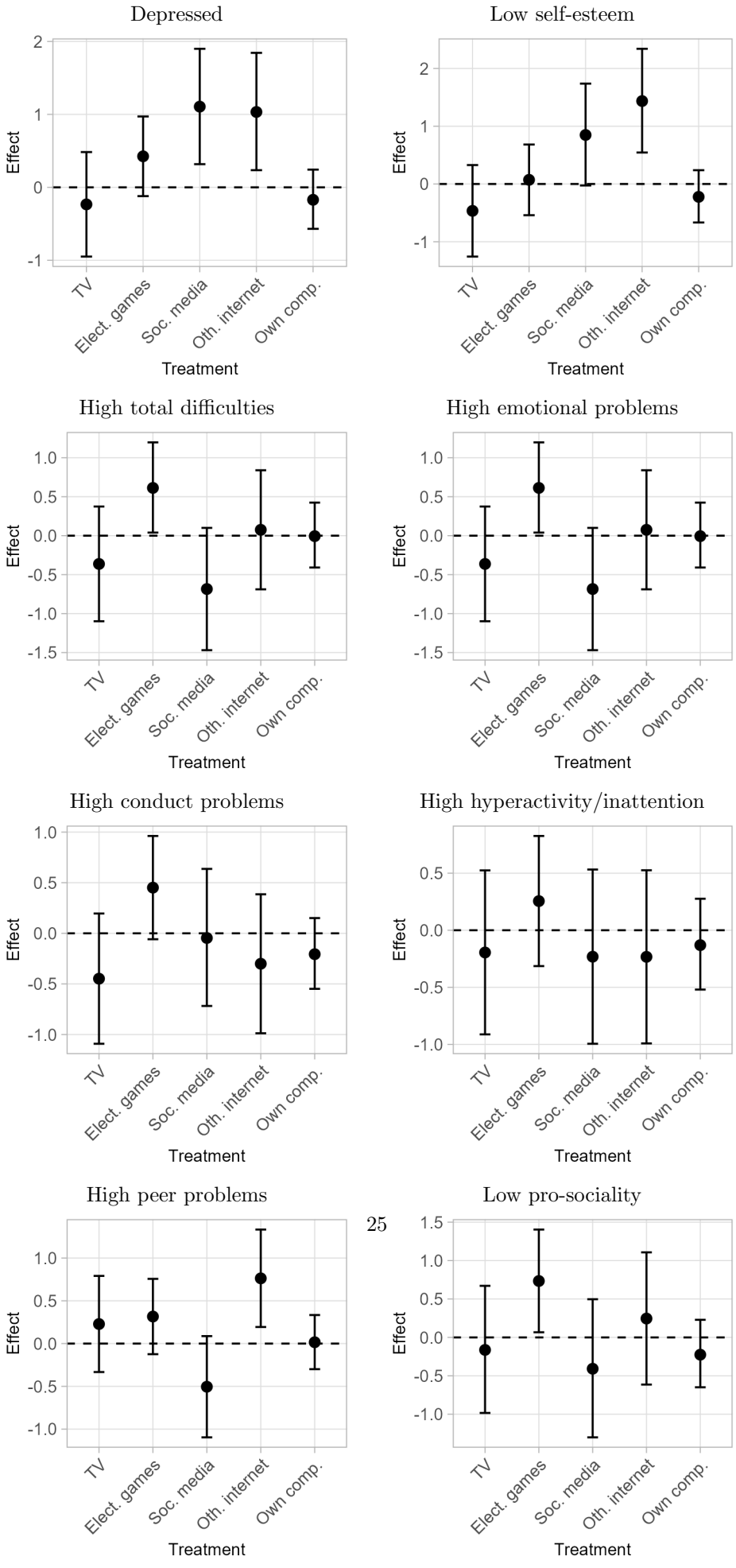


Fig. 4 Estimated treatment effects by outcome under consideration and their corresponding 95% e-confidence interval based on E^{c1}.